[3] studiați convergența și calc limita acolo unde se poate

b) $X_n = \frac{2^n}{n!}$

Monotonia — DESCRESCĂTOR

$$\frac{x^{n+1}}{x^n} = \frac{\frac{2^{n+1}}{(n+1)!}}{\frac{2^n}{n!}} = \frac{2^{n+1}}{(n+1)!} \cdot \frac{n!}{2^n} = \frac{2}{n+1} < 1, \forall n \geq 2 \Rightarrow (X_n)_{n\geq 1} \searrow \quad (*)$$

$X_n > 0 \ \forall n \in \mathbb{N} \Rightarrow (X_n)_{n\geq 1}$ - mărginit inferior $(**)$

Din $(*)$ și $(**)$ $\Rightarrow (X_n)_{n\geq 1}$ convergent $\exists l \in \mathbb{R}$ a.î. $\lim_{n\to\infty} = l$

$$\frac{x^{n+1}}{x^n} = \frac{2}{n+1} \Rightarrow X_{n+1} = X_n \cdot \frac{2}{n+1} \quad \Big/ \lim_{n\to\infty}$$

$$l = l \cdot 0 \Rightarrow l = 0$$

c) $X_n = \sqrt[n]{n}$

Fie $a_n = \sqrt[n]{n} - 1 \Rightarrow n = (1+a_n)^n = 1 + C_n^1 a_n + \underline{C_n^2 \cdot a_n^2} + \dots + a_n^n$

$$> \underline{C_n^2 \cdot a_n^2} = \frac{n(n-1)}{2} \cdot a_n^2$$

$$\Rightarrow n > \frac{n(n-1)}{2} \cdot a_n^2 \Rightarrow a_n < \sqrt{\frac{2}{n-1}}, \ \forall n \geq 2$$

$$\Rightarrow 0 < a_n < \sqrt{\frac{2}{n-1}}, n \to \infty \Rightarrow \lim_{n\to\infty} a_n = 0 \Rightarrow \lim_{n\to\infty} \sqrt[n]{n} = 1$$

d) $x_n = \left(1+\frac{1}{n}\right)^n$

$$x_n = 1 + C_n^1 \cdot \frac{1}{n} + C_n^2 \cdot \frac{1}{n^2} + \dots + C_n^n \cdot \frac{1}{n^n} =$$

$$= 1 + n \cdot \frac{1}{n} + \frac{n(n-1)}{2!} \cdot \frac{1}{n^2} + \dots + \frac{n(n-1)\cdot \dots \cdot (n-n+1)}{n!} \cdot \frac{1}{n^n} =$$

$$= 1 + 1 + \frac{1}{2!}\left(1-\frac{1}{n}\right) + \frac{1}{3!}\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right) + \dots + \frac{1}{n!}\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\dots\left(1-\frac{n-1}{n}\right)$$

$$\Rightarrow x_{n+1} = 1+1+\frac{1}{2!}\left(1-\frac{1}{n}\right) + \frac{1}{3!}\left(1-\frac{1}{n+1}\right)\left(1-\frac{2}{n+1}\right) + \dots +$$
$$+ \frac{1}{n!}\left(1-\frac{1}{n+1}\right)\left(1-\frac{2}{n+1}\right)\cdot \dots \cdot \left(1-\frac{n-1}{n+1}\right) + \frac{1}{(n+1)!}\left(1-\frac{1}{n+1}\right)\left(1-\frac{2}{n+2}\right)\dots\left(1-\frac{n}{n+1}\right)$$

$\Rightarrow x_n < x_{n+1} \quad \forall n \geq 1 \Rightarrow (x_n)$ şir cresc

$$x_n = \left(1+\frac{1}{n}\right)^n < 1+1+\frac{1}{2!}+\frac{1}{3!}+\dots+\frac{1}{n!} < 1+1+\frac{1}{2}+\frac{1}{2^2}+\dots+\frac{1}{2^{n-1}} =$$
$$= 1 + \frac{1-\frac{1}{2^n}}{1-\frac{1}{2}} = 3 - \frac{1}{2^{n-1}} < 3$$

căci $\frac{1}{n!} = \frac{1}{1\cdot 2 \cdot \dots \cdot n} < \frac{1}{\underbrace{1\cdot 2\cdot 2 \cdot \dots \cdot 2}_{n\text{-termeni}}}$ $\forall n \geq 3 \Rightarrow (x_n)$ mărginit sup

$\Rightarrow (x_n)$ convergent

e) $x_n = 1 + \frac{1}{1!} + \frac{1}{2!} + \dots + \frac{1}{n!}$

Evident că (x_n) cresc. şi mărg sup $\Rightarrow (x_n)$ convergent $\Rightarrow \exists \lim\limits_{n\to\infty} x_n = L \in \mathbb{R}$

Notăm $y_n = \left(1+\frac{1}{n}\right)^n \overset{d)}{\Longrightarrow} y_n < x_n, \forall n \geq 1 \Rightarrow e \leq L \quad (*)$

Fie $p \in \mathbb{N}^*$ fixat și $n > p$

* cu binomul lui Newton

$$\Rightarrow y_n > 1 + 1 + \frac{1}{2!}\left(1-\frac{1}{n}\right) + \dots + \frac{1}{p!}\left(1-\frac{1}{n}\right)\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\cdots\left(1-\frac{p-1}{n}\right) \quad \forall n > p,\ n \to \infty$$

$$\Rightarrow e \geq 1 + 1 + \frac{1}{2!} + \dots + \frac{1}{p!} = x_p,\ \forall p \in \mathbb{N}^*,\ p \to \infty$$

$$\Rightarrow \underline{e \geq L} \quad (**)$$

Din $(*)$ și $(**)$ $\Rightarrow \boxed{e = L}$

f) $$x_n = \frac{\sin(1!)}{1 \cdot 2} + \frac{\sin(2!)}{2 \cdot 3} + \dots + \frac{\sin(n!)}{n \cdot (n+1)}$$

* nu e monoton

* arătăm că e fundamental

(x_n) convergent $\Leftrightarrow$ (x_n) fundamental

$\Leftrightarrow \forall \varepsilon > 0, \exists \underline{n_0 \in \mathbb{N}}$ a.î. $\forall n \geq n_0, \forall p \in \mathbb{N} : |x_{n+p} - x_n| < \varepsilon$

$$|x_{n+p} - x_n| = \left|\sum_{k=1}^{n+p} \frac{\sin(k!)}{k\cdot(k+1)} - \sum_{k=1}^{n} \frac{\sin k!}{k(k+1)}\right| =$$

$$= \left|\sum_{k=n+1}^{n+p} \frac{\sin k!}{k\cdot(k+1)}\right| \leq$$

*„continuăm cu mai mic sau egal"

$|a+b| \leq |a| + |b|$

$$\leq \sum_{k=n+1}^{n+p} \frac{|\sin k!|}{k(k+1)} \leq \sum_{k=n+1}^{n+p} \frac{1}{k\cdot(k+1)} = \sum_{k=n+1}^{n+p}\left(\frac{1}{k} - \frac{1}{k+1}\right) =$$

$$= \frac{1}{n+1} - \cancel{\frac{1}{n+2}} + \cancel{\frac{1}{n+2}} - \cancel{\frac{1}{n+3}} + \dots + \frac{1}{n+p} - \frac{1}{n+p+1} = \frac{1}{n+1} - \frac{1}{n+p+1}$$

$$< \frac{1}{n+1} \underset{\substack{\uparrow \\ \text{punem}}}{<} \varepsilon$$

$\Rightarrow n+1 > \frac{1}{\varepsilon} \Rightarrow n > \frac{1}{\varepsilon} - 1$, alegem $n_0 = \left[\frac{1}{\varepsilon}\right]$

$n \geq n_0 > \frac{1}{\varepsilon} - 1$

Suplimentare

$\boxed{3}$

$x_n = 1 + \frac{1}{1!} + \frac{1}{2!} + \dots + \frac{1}{n!}$, $y_n = x_n + \frac{1}{n \cdot n!}$

a) (y_n) - strict descresc şi $\lim\limits_{n\to\infty} x_n = e$

b) $0 < e - x_n < \frac{1}{n \cdot n!}$ $\forall n \in \mathbb{N}^*$

a)

$$y_{n+1} - y_n = \cancel{x_n} + \frac{1}{(n+1)!} + \frac{1}{(n+1)(n+1)!} - \cancel{x_n} - \frac{1}{n \cdot n!} =$$

$$= \frac{1}{(n+1)!} + \frac{1}{(n+1)(n+1)!} - \frac{1}{n \cdot n!} =$$

$$= \frac{n(n+1) + n - (n+1)^2}{n(n+1)(n+1)!} = \dots = \frac{-1}{n(n+1)(n+1)!} < 0 \Rightarrow y_n \searrow$$

$$y_n = x_n + \frac{1}{n \cdot n!} \Rightarrow \lim_{n\to\infty} y_n = e + 0 = e$$

b) $0 < e - x_n < \frac{1}{n \cdot n!} \iff \begin{cases} x_n < e, & (x_n \nearrow) \\ e < y_n, \forall n \geq 1 & (y_n \searrow) \end{cases}$

c) Presupunem prin absurd:

că $e = \frac{p}{q} \in \mathbb{Q}$, $p, q \in \mathbb{N}^*$

$$\overset{b)}{\Longrightarrow} 0 < \frac{p}{q} - x_n < \frac{1}{n \cdot n!}, \quad \forall n \geq 1$$

$$n = q \Rightarrow 0 < \frac{p}{q} - x_q < \frac{1}{q \cdot q!} \quad \Big| \cdot q \cdot q!$$

$$\Rightarrow 0 < \underbrace{p \cdot q! - q \cdot q! \cdot \left(1 + \frac{1}{1!} + \frac{1}{2!} + \ldots + \frac{1}{q!}\right)}_{\in \mathbb{Z}} < 1 \text{ - absurd}$$

$\Rightarrow$

$$\Rightarrow e \notin \mathbb{Q}$$